

This exam consists of 19 questions.

Points are indicated for each question and broken down to each part.

Question 1: Software Crisis (4 points)

Explain the term Software Crisis and list commonly mentioned reasons for it.

Question 2: Software Engineering Phases (5 points)

List five traditional software engineering phases. Briefly describe each in one short sentence.

Question 3: RUP (6 points)

Explain the Rational Unified Process based on its phases and workflows.

Question 4: Lifecycle Model Choice (6 points)

List 3 classic software engineering mistakes and explain how the Spiral model addresses them.

Question 5: Stakeholders (4 points)

Name four stakeholders for a software for managing patients in a hospital and explain their relevant concerns for the software.

Question 6: Requirements classification (5 points)

For each of the following requirements, mark if they are functional (F) or non-functional (N):

- (a) ___ The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.
- (b) ___ Downtime within normal working hours shall not exceed 5 seconds in any one day.
- (c) ___ A user shall be able to search the appointments lists for all clinics.
- (d) ___ The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
- (e) ___ Each staff member using the system shall be uniquely identified by their eight-digit employee number.

Question 7: User stories (4 points)

Write two user stories as requirements for an online visa application system.

Question 8: Traceability (4 points)

What is a traceability matrix and what is it used for.

Question 9: Feature interaction (4 points)

Define feature interaction and explain two interaction types.

Question 10: Model criteria (6 points)

List three model criteria and illustrate them on an example model.

Question 11: Programming language selection (6 points)

List three selection criteria for a programming language and illustrate each on an example.

Question 12: Bugs (3 points)

Define the terms failure, fault, and error in the context of software testing.

Question 13: Inspection vs. testing (3 points)

Give an example where inspection could be used instead of testing and explain why.

Question 14: Testing techniques (4 points)

Contrast exhaustive testing and combinatorial test design.

Question 15: Combinatorial test design (8 points)

Use the In-Parameter-Order algorithm to complete the test plan below for pairwise coverage ($t = 2$).

The parameters are $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, and $C = \{c_1, c_2, c_3\}$.

a_1 b_1

a_1 b_2

a_2 b_1

a_2 b_2

Question 16: Lines of Code (4 points)

Name one reason why lines of code is useful and one reason why it is not useful as a metric.

Question 17: White-box Testing

- (a) (6 points) Draw the control flow graph for the given code snippet (the lines are spaced to draw the graph next to the code). Labeling nodes with the line number is sufficient.

```
1 public int myfunction(int a, int b) {  
2     int d = a + b;  
3     if (a < b) {  
4         a = 2 * a;  
5         if (a > b) {  
6             return a;  
7         }  
8     }  
9     int i = 0;  
10    while (i < d) {  
11        a++;  
12        i++;  
13    }  
14    return a + b;  
15 }
```

- (b) (2 points) Provide a minimal set of test cases for 100% statement coverage.

- (c) (3 points) Explain the difference between branch coverage and statement coverage. Provide a minimal set of test cases for 100% branch coverage.

(d) (2 points) Explain the relation between path coverage and condition coverage.

(e) (2 points) Compute the cyclomatic complexity of the program.

Question 18: SCRUM (5 points)

Describe the high-level process of SCRUM.

Question 19: Code smells (4 points)

Explain the concept of code smells and give two examples.

(This is an additional sheet for your answers.)

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